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Integrating deep learning and data fusion for advanced keystroke dynamics authentication

Arnoldas Budžys *, Olga Kurasova, Viktor Medvedev

Institute of Data Science and Digital Technologies, Vilnius University, Akademijos str. 4, Vilnius 08412, Lithuania

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ABSTRACT

By enhancing user authentication protocols, especially in critical infrastructures vulnerable to complex cyberthreats, we present an advanced approach that integrates a deep learning-based model and data fusion techniques applied to analyze keystroke dynamics. With the growing need for robust security measures, especially in critical infrastructure environments, traditional authentication mechanisms often fail to cope with advanced threats. Our approach focuses on the unique behavioral biometric characteristics of keystrokes, which offers promising opportunities to improve user authentication processes. We have developed a data fusion-based methodology that utilizes the unique features of keystroke dynamics combined with deep learning techniques to improve user authentication systems. Using the capabilities of data fusion and deep learning, the proposed methodology not only captures the complex behavioral biometrics inherent in keystroke dynamics but also addresses the challenges posed by varying password lengths and typing styles. We conducted extensive experiments on several fixed-text datasets, including the Carnegie Mellon University dataset, the KeyRecs dataset, and the GREYC-NISLAB dataset, with a total of approximately 54,000 password records. Comprehensive experiments on various datasets with different password lengths have shown that our approach is scalable and accurate for user authentication, which significantly improves the security of critical infrastructure. By using interpolation-based data fusion techniques to standardize the keystroke data to a uniform length and employing a Siamese neural network with a triplet loss function, the best equal error rate of 0.13281 was achieved for the unseen fused data. The integration of deep learning and data fusion effectively generalizes to different user profiles, demonstrating its adaptability and accuracy in authenticating users in different scenarios. The findings are crucial for improving security in sensitive applications, ranging from accessing personal devices to protecting critical infrastructure.

1. Introduction

Cybersecurity threats are becoming increasingly sophisticated, making traditional authentication methods such as passwords increasingly vulnerable. This research aims to address these threats by implementing a more secure authentication method. The importance of robust cybersecurity measures, such as the use of deep learning to analyze keystroke dynamics in user authentication, cannot be overstated. Multi-factor authentication is increasingly being recognized as a critical component of cybersecurity strategies across a variety of sectors, including banking, healthcare, retail, and Internet of Things devices. The importance of multi-factor authentication is also highlighted by the fact that 80% of hacker data breaches are due to weak or compromised passwords, making the improvement of authentication methods paramount. The use of deep learning in this area is particularly promising. Unlike traditional statistical machine learning methods, deep learning offers advanced pattern recognition capabilities, making it more effective in analyzing complex typing behaviors. The goal of this research is to utilize these capabilities to develop more sophisticated and secure methods of authentication, which is crucial for protecting critical infrastructure systems.

Cybersecurity is particularly important for critical infrastructures, as they are vital for public safety, economic prosperity, and national security, and their vulnerability is increasing due to the growing risk of cyberthreats. This requires a specific approach to cybersecurity to ensure that essential services are up and running and protected from cyberattacks. The Cybersecurity and Infrastructure Security Agency (CISA) [1] identifies 16 critical infrastructure sectors whose assets, systems, and networks are integral to national security, health care, and public health. These sectors, ranging from the defense industrial base sector and nuclear reactors to healthcare and financial services,

* Corresponding author.
E-mail addresses: arnoldas.budzys@ieee.org (A. Budžys), olga.kurasova@mif.vu.lt (O. Kurasova), viktor.medvedev@mif.vu.lt (V. Medvedev).

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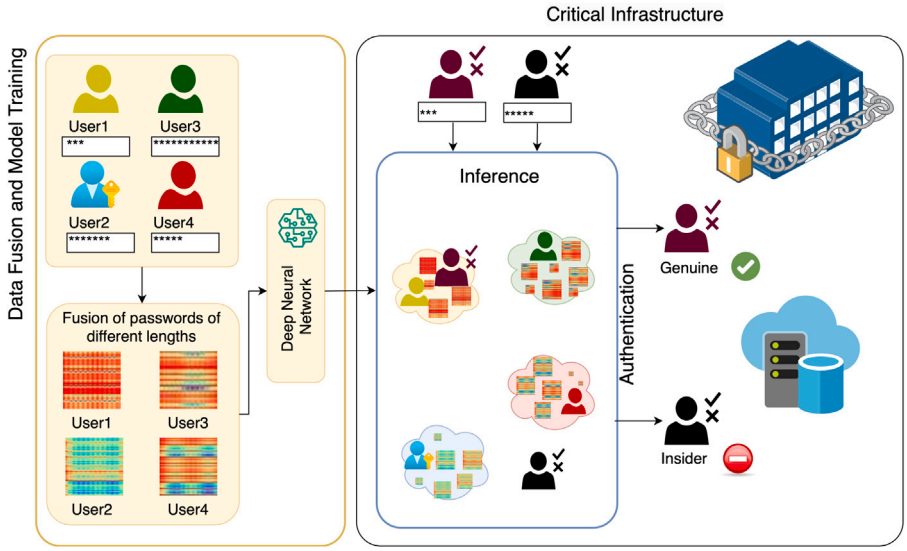

Fig. 1. Schematic representation of the keystroke dynamics-based authentication system for critical infrastructure, illustrating the process from data collection to user authentication using deep learning. The system combines multiple datasets, applies feature extraction and data fusion, and uses a deep neural network for final user authentication.

highlight the wide range of vulnerabilities and the need for comprehensive cybersecurity solutions. In environments where the keyboard is the primary interface, such as data centers, military installations, or other critical facilities, authenticating the identity of users is not only a matter of protecting data but also of protecting critical infrastructures. In addition, as the World Economic Forum emphasizes [2], protecting systemically important critical infrastructures is vital for preventing disruptions that could have serious consequences. This underscores the importance of cybersecurity measures, including secure user authentication processes, to prevent unauthorized access and reduce the risk of cyberattacks. The current focus of research and discussion in the field of information security is the development of novel authentication techniques, particularly those that leverage keystroke dynamics to authenticate users based on their typing behavior. These techniques are aimed at preventing security breaches in critical infrastructure. To improve the efficiency of these authentication processes in critical infrastructure applications, we use data fusion techniques to standardize the keystroke data to a uniform length and train a deep learning-based model on diverse datasets. This enables the use of passwords of different lengths and provides strong verification of users across different profiles and scenarios. Fig. 1 schematically illustrates the keystroke dynamics authentication process, demonstrating conceptually how the proposed methodology can be applied to user authentication in the described context. Keystroke biometrics is a form of behavioral authentication that verifies users by analyzing the unique biometric patterns of their typing behavior. The implementation of this authentication system in sensitive sectors, such as critical infrastructure, significantly enhances the efficacy of defenses against insider threats. Even if an insider manages to acquire a user's password by observing over the shoulder or using software to capture it, they will not be able to replicate the genuine user's unique typing behavior. Consequently, when the insider attempts to access the system, the disparity in typing patterns prevents them from successfully connecting, thereby mitigating potential harm.

A study of keystroke dynamics for biometric authentication [3,4] revealed a difference between fixed-text and free-text biometric systems. While fixed-text scenarios show higher accuracy, the performance of free-text algorithms lags behind significantly, especially due to the variability in free-text input. This variability is a significant challenge in recognizing subjects based on keystroke dynamics, highlighting the need for deep learning-based models that can efficiently navigate the complexities of free-text scenarios or different text lengths, e.g., different passwords. Despite its advantages, such as ease of data collection and the ability to monitor passively, keystroke dynamics faces challenges, including variability in typing patterns due to emotional state or different keyboards [5]. Our study aims to address these challenges by using advanced deep learning techniques that are able to generalize across different typing behaviors and adapt to user-specific patterns.

Users typically use passwords of different lengths, so each password results in a time series of different lengths, which are generated from features that correspond to the key press times. This variability in password length requires authentication systems to have specific solutions that can efficiently handle different lengths of input data. In such scenarios, data fusion becomes necessary as it allows the lengths of different passwords to be unified and different datasets to be fused so that a deep learning model can be trained on diverse datasets. This approach ensures that the model adapts well to the variability of fixed-text input while maintaining adaptability to different samples of passwords. Using data interpolation techniques, we can create a fused dataset that captures patterns of keystroke dynamics associated with different password lengths. This fused dataset is the basis for training the Siamese neural network, enabling detailed analysis and identification of insider threats in the context of cybersecurity.

Thus, this research aims to contribute to the field of cybersecurity by developing a novel approach to analyze keystroke dynamics for user authentication based on deep learning and data fusion, with a particular focus on its application to critical infrastructure protection. The proposed methodology and the obtained results can significantly improve security for critical systems, offering a solution that is both secure and adaptable to the changing cyberthreat landscape. The application of deep learning to keystroke dynamics is constrained by the scale of the data and the diversity of devices. Most existing studies are limited to a few hundred subjects, which prevents the full potential of deep learning to extract knowledge from large-scale datasets. The research proposed in this paper aims to fill this gap by utilizing deep learning to scale keystroke biometric recognition to larger populations and a greater diversity of devices.

The contributions of this work are as follows:

- A methodology for data fusion-based authentication is proposed that standardizes keystroke dynamics data from different datasets. This allows the creation of a deep learning-based model capable of handling passwords of different lengths, thereby significantly enhancing the generalizability and robustness of user authentication systems. Moreover, this methodology ensures compatibility with passwords from short to long lengths, effectively addressing the crucial challenge within behavioral biometric authentication.

- Two categories of interpolation-based data fusion methods are applied: one to direct keystroke discrete signals and the other for image resizing using interpolation techniques. Our results enable us to identify the most effective strategies for processing variable-length passwords, thereby improving the accuracy and flexibility of user authentication systems across different profiles and scenarios, including critical infrastructure applications.
- This research introduces a novel approach to keystroke dynamics by integrating and standardizing different datasets — specifically, the Carnegie Mellon University dataset, the KeyRecs dataset, and the GREYC-NISLAB dataset — into a unified format suitable for deep learning models. This approach, which has not been previously explored in the literature, involves combining diverse datasets, encompassing approximately 54,000 password records from a wide demographic spectrum, to train a deep learning model for adaptive user authentication.

The remainder of the paper is structured as follows. Section 2 provides a summary of related work on fixed-text keystroke dynamics in the context of user authentication. Section 3 proposes a methodology for user authentication to the system. The experimental setup and the obtained results are presented and discussed in Section 4. Finally, Section 6 concludes the paper by highlighting the main findings and discussing possible directions for future research.

2. Related works

In the emerging field of cybersecurity, keystroke biometrics has become a sophisticated tool for user identification and authentication based on the inherent typing patterns and behaviors of individuals [6]. The article [7] highlights the probabilistic nature of biometric systems, such as keystroke dynamics, highlighting the difficulty of accurately authenticating a user due to variables such as typing behavior and emotional factors. Comprehensive research and a systematic approach to enhancing the robustness and real-world applicability of biometric identification technologies are needed. The keystroke dynamics, consisting of features such as typing speed, rhythm, and key pressure, create a unique biometric profile. This profile enhances security when combined with established authentication methods such as alphanumeric passwords or physiological biometrics. The keystroke dynamics authentication system requires no intervention, is cost-effective, and can be used in a variety of environments such as computers, mobile devices, ATMs, and, in particular, critical infrastructure systems. In addition, the way a person type is difficult to reproduce, so this method is more effective in identifying insiders [3]. The advantage of the keystroke dynamics authentication system is that users are not required to perform specific tasks, as this method runs continuously in the background [8]. Despite their advantages, keystroke dynamics authentication systems are limited by the variability of individual typing behavior. This variability can be influenced by temporary conditions such as tiredness or an emotional state. Factors such as physical disabilities or the use of assistive typing devices may also affect writing patterns. In addition, the potential variability of typing patterns presents a challenge; sophisticated fraudsters may attempt to replicate the keystroke dynamics of another user, with varying degrees of success.

Deep learning has contributed significantly to various fields, including cybersecurity [4,9]. Its ability to process and learn from large amounts of data, recognize patterns, and make predictions has made it a powerful tool for improving the effectiveness of cybersecurity measures. Deep learning algorithms are particularly good at extracting complex, high-dimensional characteristics from raw biometric data such as fingerprints, iris scans, and voice signals. This allows for more accurate and detailed recognition than traditional algorithms. Biometric characteristics can change due to many factors, including age, environment, and physical condition. Deep learning models can adapt to these changes, ensuring stable performance of the authentication system over time. In addition, they can be trained to distinguish genuine biometric features from synthetic copies or spoofed characteristics, increasing the security of biometric systems against fraudulent access attempts. By integrating data from multiple biometric sources, deep learning can facilitate the development of multimodal biometric systems. These systems improve the accuracy of user recognition or authentication by using a variety of biometric features. The paper [10] introduced TypeNet, a deep learning-based approach for keystroke biometrics that achieves state-of-the-art performance by training on specific datasets tailored for fixed-text scenarios.

Identity, authentication, and access control management are pivotal for ensuring that only authorized users have access to specific assets and can perform approved actions. The deployment of deep learning or machine learning techniques enhances user authentication through physical or behavioral biometrics, including multi-factor authentication [11–13]. Machine learning algorithms, which are widely utilized in this field, have been applied to various datasets, and have often achieved significant improvements in user authentication accuracy. Notably, research using machine learning for keystroke dynamics has reported enhancements in multi-user identification accuracy of up to approximately 94% [14–16]. Keystroke biometrics, a crucial method in user behavioral biometrics authentication, encompasses both static and dynamic categories. Both static and dynamic keystroke biometric authentication involve the collection of keystroke patterns and the extraction of features such as timing and press patterns. Static authentication utilizes a predefined sequence of keystrokes, typically in controlled input scenarios such as entering a username or password. Conversely, dynamic or continuous authentication operates with any keystroke sequence, is applicable in unstructured text environments such as emails or social media, and continuously verifies the user's identity throughout a session. This method builds a unique keystroke profile for each user, reflecting their specific rhythmic and temporal typing patterns, which are essential for the authentication process. In addition, advanced techniques such as a single multi-class Convolutional Neural Network (CNN) model trained with specific data augmentation methods have significantly reduced the Equal Error Rate (EER) to 0.023, demonstrating the efficacy of these approaches in reducing authentication errors [17], while without augmentation, the EER was 0.065. Research on keystroke-based authentication, particularly for critical infrastructures, employs various analytical methods across both static and dynamic contexts. Techniques such as Dynamic Time Warping (DTW), Manhattan distance, and statistical models, including hidden Markov models, are employed not only in static authentication scenarios but also in continuous/dynamic authentication, to compare keystroke rhythms and speeds against a trained model for robust user verification [18–20]. A recent study by [21] introduced a multi-user identification algorithm for fixed-text keystroke dynamics, employing advanced clustering and dimensionality reduction techniques for high accuracy in user distinction.

The most widely used dataset in the field of keystroke dynamics for static authentication is the dataset proposed by Carnegie Mellon University (CMU) [22]. The purpose of the papers (see [22–24]) was to achieve the lowest possible EER for the CMU data using different methods. The Manhattan distance (scaled) method reported by [22] achieved an EER of 0.096. The nearest neighbor approach with the new distance metric combining the Mahalanobis distance and the Manhattan distance, examined by [25], resulted in an EER of 0.087 on the validation dataset, and after removing the outliers, it resulted in an EER of 0.084 on the validation dataset. The inductive transfer encoder approach, as utilized by Monaco [26], resulted in an EER of 0.063, which was closer to the result of the CNN used by [17], with an EER of 0.065 on the validation dataset. In addition, methods such as dependency clustering with Manhattan distance [27] and Manhattan distance (scaled with standard deviation) [28] showed EERs of 0.077 and 0.0916, respectively. These abovementioned results for the CMU

dataset are obtained on validation data or by eliminating outliers, which is irrational in a real-life scenario, as behavioral biometrics depend on a person's emotional or physical state.

Additional relevant datasets such as KeyRecs [29] and GREYC-NISLAB [23] are analyzed in depth in [29,30]. The equal error rate is a key metric in the evaluation of authentication systems for behavioral biometrics, reflecting a trade-off point where the false rejection rate associated with legitimate samples exactly matches the false acceptance rate associated with insider samples. This metric serves as a critical indicator of a system's authentication accuracy, balancing security and usability.

New research in this area has focused on the application of artificial intelligence-based techniques. The DeepSecure method described in [31] demonstrated significant progress by deploying a four-layer perceptron architecture trained individually for each user. Another study [17] used a single multi-class convolutional neural network with a novel data augmentation strategy, thereby demonstrating the effectiveness of neural network approaches in improving authentication accuracy. User authentication based on keystroke dynamics emphasizes the effectiveness of deep learning algorithms.

Siamese neural networks (SNNs) [32] have become widely used in the field of biometric authentication, going beyond their traditional applications to offer an advanced approach for learning subtle differences in data similarity. An SNN is a special type of neural network architecture consisting of two or three identical branches of sub-networks that share the same parameters and are trained together. These branches process different input data but are optimized to produce similar or correlated output data. Such a method is particularly useful in biometric authentication, as it effectively addresses class imbalances in intrusion detection by providing accurate similarity metrics between biometric data points. The basic idea of an SNN is to learn a similarity or distance metric between inputs by comparing the output representations of the branches. SNNs first use a contrastive loss function during training [33], which encourages the network to produce similar representations for similar inputs and dissimilar representations for dissimilar inputs. This helps the network to learn a reliable similarity metric between inputs. In recent years, researchers have started using SNNs for intrusion detection systems [34–36]. The FaceNet [37] method uses a triplet loss function for training SNN with CNN branches. Its goal is to preserve the difference between each pair of faces belonging to one person and all other faces. A triplet loss function [37] is a type of loss function that is commonly used in training SNNs to obtain similar representations for similar input data and dissimilar representations for dissimilar input data. Triplets consisting of anchor, positive, and negative representations are required to train the network. The comparison is performed in latent space. This solution can also be used for biometric identification when the network receives two related samples (anchor, positive) and one different sample (negative).

The study [38] addresses the issues of enhancing computer and certain infrastructural security through advanced user authentication methods, in particular the potential of keystroke dynamics as a cost-effective alternative to traditional physiological biometric systems. Biometric authentication methods such as fingerprints, voice recognition, and retinal scans require specialized sensors and can be invasive, which creates problems with user privacy and implementation costs. Compared to physiological biometrics, behavioral biometrics, especially keystroke dynamics, offers a less invasive and more cost-effective authentication solution.

Behavioral biometrics, especially keystroke dynamics, collects time series data, and each password or passphrase entered has a unique set of features. In the case of time series analysis, fusing different datasets into a single dataset is a challenging task that can be addressed by a variety of approaches. Interpolation techniques are particularly effective in standardizing datasets of different lengths to make them more uniform for analysis. Statistical or machine learning-based methods, such as cubic, linear [39], nearest neighbor [40], resampling, DTW [18], and Fourier transform, can be used to unify time series of different lengths. These methods can be used to standardize datasets and thus facilitate the application of complex analytical models. When a network is trained on the fused dataset, its generalizability is enhanced, allowing it to perform more effectively across different types of data in real-world scenarios. DTW is an efficient method for aligning sequences that differ in speed or time; it requires significant computational resources and is less appropriate for larger datasets. This is because the process of determining the alignment of two time series is complex (typically $O(n \times m)$, where n and m are the lengths of the two sequences being compared). The Fourier transform is less efficient than simpler interpolation methods, especially for large datasets or applications that require real-time processing. In contrast, interpolation methods such as linear, cubic, and nearest neighbor methods are much less computationally intensive. Each interpolation method has its own advantages and limitations, so the choice of method depends on the specific characteristics of the data to be analyzed and the desired result of the interpolation process.

3. Methodology

The rapidly evolving cybersecurity landscape makes it necessary to strengthen the defenses of critical infrastructure systems to reduce vulnerabilities and the risk of unauthorized insider access. Such individuals can obtain system administrator credentials by surreptitiously monitoring the system or by using software tools designed to extract passwords. Once these credentials are obtained, an insider can cause damage to a critical infrastructure system by impersonating another user, thereby covering their tracks. Therefore, there is a growing need for advanced Intrusion Prevention Systems (IPS) that utilize deep neural networks to investigate biometric patterns of user typing behavior. Static authentication is a method of user authentication that requires a password to gain access to the system. Each user has a unique typing behavior, in particular in the way they enter their password. When a user enters a password, the system captures timestamps related to their keystroke dynamics, including the hold time, release-to-press time, press-to-press time, and release-to-release time. These timestamps allow us to analyze individual characteristics, providing insights into the unique typing patterns of each user (see more details in [9,41]).

In our previous researches [9,41,42], we presented a novel user authentication methodology that uses a deep learning-based SNN with CNN branches and a triplet loss function to authenticate users on their unique password typing behavior. The novelty of this methodology lies in its ability to learn and distinguish individuals' typing patterns. Transforming keystroke dynamics data into images takes advantage of the proven mathematical ability of CNNs to extract sophisticated features. This network effectively uses filters of varying sizes to capture and analyze data at different spatial resolutions, thereby enhancing the system's ability to learn and recognize typing patterns efficiently. This approach exploits the strengths of CNNs, allowing the development of more accurate and reliable user authentication systems based on unique typing behavior. We have proposed a novel method for transforming non-image data into images named GAFMAT (GAbor Filter MAtrix Transformation) [42]. This method leverages CNN's advanced capabilities to extract features from the images for higher accuracy and lower EER for user authentication systems.

3.1. Data fusion-based methodology for user authentication

In this paper, we further develop the methodology of user authentication based on keystroke dynamics. The main novelty lies in the fusion of several different datasets. Data fusion contributes to the generalized learning capability of the deep learning model, making it resilient to the variability inherent in real-world authentication scenarios.

The data fusion-based authentication methodology utilizing complex keystroke dynamics involves two steps: determining a suitable

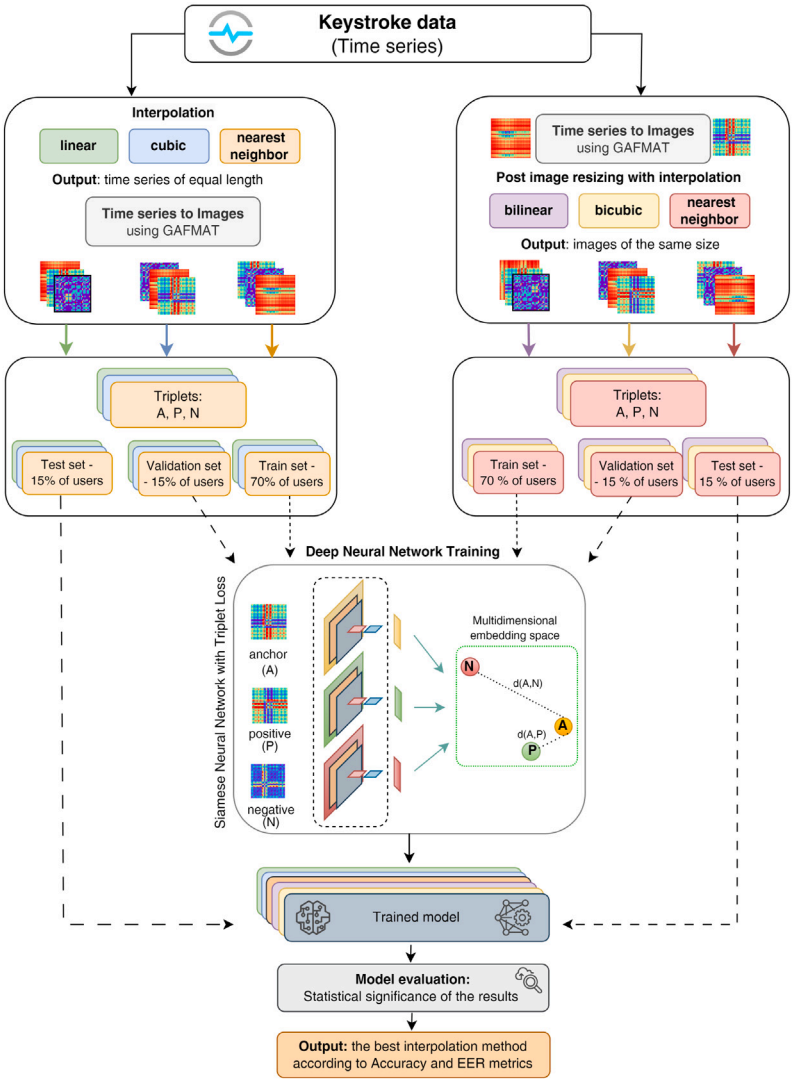

Fig. 2. Flowchart for selecting the most appropriate interpolation method for data fusion to unify keystroke dynamics data from multiple datasets: the decision-making process for applying interpolation methods to either original time series data or images transformed using the GAFMAT approach, resulting in a unified data format for Siamese neural network training.

interpolation method (see Fig. 2) and implementing data fusion-based authentication using complex keystroke dynamics (see Fig. 3).

The flowchart in Fig. 2 illustrates the methodology for selecting the most appropriate interpolation method to unify keystroke dynamics data from different datasets. Three interpolation methods (linear, cubic, and nearest neighbor) are first applied to the original time series to standardize the number of features of each password (see the left side of Fig. 2). After these interpolation processes, the GAFMAT approach (see Section 3.2) is used to transform the output into images, effectively converting passwords of different lengths into images of the same dimensions. Another possible way to unify the dimensions of the data is to first transform the time series corresponding to the passwords into images using the GAFMAT method, and then unify the size of the resulting images using interpolation techniques such as bilinear, bicubic, and nearest neighbor methods (see the right side of Fig. 2). Details on time series interpolation and image resizing with interpolation are described in Section 3.3. Regardless of which data standardization method is used, the resulting images are used to form triplets—groups consisting of an anchor, a positive sample from the same user, and a negative sample from another user. These triplets are used to train an SNN to distinguish the typing behavior of the keystroke dynamics of the users. The data are divided into separate datasets for training (70% of users), validation (15% of users), and testing (15% of users). This splitting allows for the evaluation of the predictive performance of the network and its ability to generalize to unseen data. The trained network embeds the data in a multidimensional space, where the distances between anchor-positive and anchor-negative pairs are analyzed. The goal of the model is to minimize the distance between similar (anchor and positive) pairs and maximize the distance between dissimilar (anchor and negative) pairs. After training, the model is evaluated by examining the accuracy and EER. The evaluation included calculating the statistical significance of the results to ensure the reliability of the conclusions (see Section 4.2). Therefore, based on the accuracy and EER, the best interpolation method is determined that best standardizes the keystroke dynamics data for the deep learning model.

The methodology presented in Fig. 3 is designed to authenticate users by analyzing the unique characteristics of keystroke dynamics. It establishes a systematic approach for fusing all passwords for user identification. Initially, the CMU, KeyRecs, and GREYC-NISLAB (LDC, MS, RHCP, TRS, USA) datasets, which vary in the number of users and features, are standardized by interpolating to a single number of features to ensure uniformity before further processing. Details on the datasets are provided in Section 3.4. By fusing data from multiple datasets to a unified size, the approach ensures consistent input data

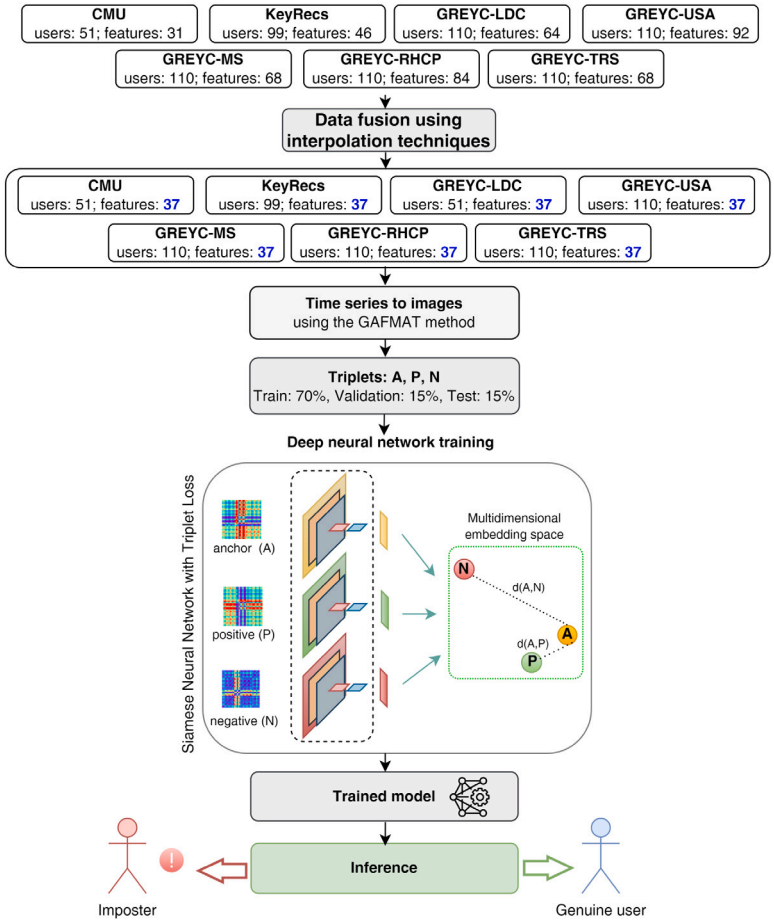

Fig. 3. Methodology for data fusion-based authentication using complex keystroke dynamics analysis. It includes steps for standardizing datasets through interpolation, transforming password samples into images, and using a trained Siamese neural network to compare embeddings of new inputs with stored records for user authentication.

for an SNN, allowing the model to be trained on a diverse set of user input patterns. In the inference phase, the user provides a sample of his/her password, which is then transformed into an image. The trained network processes this image and generates an embedding, the vector representation of the image in multidimensional space. The embedding is then compared to the embeddings of previously stored password images of the same user. The distance between the embeddings in multidimensional space is computed for this comparison. A very important part of this methodology is the establishment of a threshold to distinguish between genuine and insider attempts. This threshold is initially determined from the historical data of legitimate logins, taking into account the distance range found in the embeddings of real password records. An attempt is considered genuine if the distance does not exceed a specified threshold; otherwise, it is considered an insider.

3.2. GAFMAT approach

The GAFMAT (GAbor Filter MAtrix Transformation) approach, proposed in previous studies [9,41,42], is an advanced process for transforming time series data extracted from keystroke dynamics into images, as shown in Fig. 4, thereby improving the accuracy of authentication systems. When entering a password or passphrase, a sequential dataset is generated that captures each keystroke dynamics timestamps such as the hold time, release-press time, press-press time, and release-release time. This time series forms a discrete signal a (as shown in the upper left corner of Fig. 4), representing the unique rhythm and speed of the user's typing behavior.

The next step applies the Gabor filter [43], a well-known feature extraction method in image processing adapted for univariate time series data. The filter, which is mathematically expressed and graphically illustrated in the central part of Fig. 4, is tuned appropriately to highlight significant features in keystroke data. The idea behind this method, shown in Fig. 4, is to apply a Gabor filter to the discrete signal obtained from the keystroke dynamics. The Gabor filters, whose parameters are the filter width (σ), orientation (θ), wavelength (λ), and phase offset (ψ), are taken from given values to emphasize key features in the data, resulting in the a signal being converted to a b signal. Tuning the Gabor filter involves a sequence of operations in which the filter is applied repeatedly, each time with a different set of unique parameters. For example, σ takes values $\{2, 4, 8, 16\}$, θ is set to values within $\{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$, λ is varied among $\{16, 8, 4, 2\}$, and ψ takes the values $\{0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}\}$. The Gabor filter helps to extract certain characteristics from keystroke signals, thereby improving the dynamic analysis of keystrokes. As a result of the filters, the b signal becomes a transformed series enriched with distinctive features emphasized by applying the Gabor filter.

Subsequently, this filtered time series, together with the original time series of keystroke dynamics, is used to create a two-dimensional image ("image2D" in Fig. 4) using an outer product operation. This converts a sequence of discrete signals as a time series into a two-dimensional format that can be visually represented. In this case, the variations and patterns in the timing of keystrokes (which are captured sequentially as they occur over time) are translated into an image, where these temporal patterns are represented as variations in color or intensity patterns over the entire image.

The result is an image that represents typing behaviors in a visually interpretable format (as shown at the bottom of Fig. 4). This two-dimensional image not only depicts the complexity of keystroke

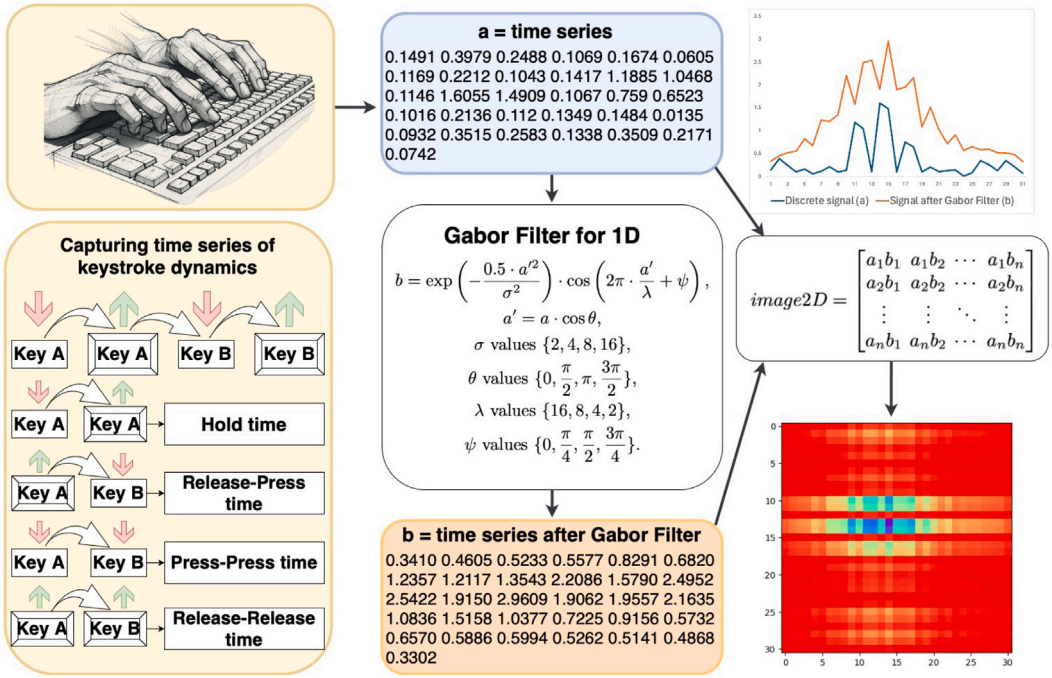

Fig. 4. The GAFMAT approach for transforming keystroke dynamics time series into two-dimensional images. The process illustrates the application of Gabor filters to emphasize significant features in the data, followed by transforming the filtered data into a two-dimensional image that represents typing behavior. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

dynamics as a visual pattern but also provides a basis for subsequent pattern recognition and deep learning analysis. Our approach demonstrates a novel approach for transforming keystroke dynamics into a format that is visually informative. Furthermore, such transformation is particularly important for training SNNs, as it provides an image that reflects the user's typing behavior. The resulting images allow the SNN to learn efficiently and distinguish genuine users from insiders.

3.3. Interpolation-based data fusion

Commonly, SNNs, like other neural networks, are trained on a single dataset, which is limited by the unique feature size. In the context of user authentication based on keystroke dynamics, passwords collected from different datasets may exhibit different lengths. When combining password datasets of different lengths, a method is needed to ensure that the format of the data fed into the SNN is uniform. In our study on keystroke dynamics for user authentication, the analyzed data are time series in which each point is a timestamp. These timestamps reflect the unique rhythm and timing of typing, which are key factors in authenticating a user's identity. In our analysis, "features" specifically refer to the timestamps associated with each keystroke, providing a detailed temporal pattern of password entry. These detailed temporal features are essential for distinguishing between individual users, which improves the accuracy of the authentication process.

Given that an SNN requires a constant size of input data, it is essential to standardize the length of the time series (the number of features) across different datasets. Various interpolation methods have been proposed to address this challenge [44]. While these methods are mainly used to fill in missing values in the time series, they can also be used to efficiently adapt the length of the time series to a uniform scale [45]. For interpolating time series data, three common interpolation methods can be applied:

- In linear interpolation, new data points in a discrete set of known data points (timestamps) are generated by assuming a straight-line progression between points. This method calculates intermediate values by connecting each pair of adjacent data points with a straight line, effectively standardizing the length of the time series by filling in or expanding the data to a uniform scale.
- In cubic interpolation, a smoother and more continuous curve is constructed through known data points using the values of neighboring points. This approach, in which a cubic polynomial is placed between each pair of data points, provides a greater degree of smoothness and accuracy than linear interpolation, resulting in a more accurate representation of the original time series.
- Nearest neighbor interpolation assigns the value of the nearest data point to any new data point, effectively preserving the characteristics of the original dataset.

An alternative approach for standardizing data dimensions involves resizing images using various interpolation methods:

- The bilinear method, which is a two-dimensional extension of linear interpolation, averages the nearest 2×2 grid of pixels, balancing efficiency and image quality with moderate scaling.
- The bicubic method, a more complex method, uses a 4×4 nearest pixel grid and cubic polynomials for a smoother transition, resulting in higher quality images that are ideal for significant resizing.
- The nearest neighbor method, which is the simplest and fastest, directly assigns the nearest pixel value to each new pixel and preserves the original characteristics of the images.

3.4. Datasets

In order to demonstrate the effectiveness of the proposed methodology, experiments were carried out using three publicly available fixed-text datasets:

- The CMU dataset [22] consists of 51 participants, each of whom was instructed to enter the password ".tie5Roanl" over eight sessions. The number of timestamps (features) is equal to 31. During each session, participants were required to enter the password 50 times, resulting in a complete dataset of 20,400 password records.

- The KeyRecs dataset [29] consists of 99 participants from all over the world, each performing the "vpwjkeurkb" password task structured dataset into 19,773 samples. The number of timestamps (features) is equal to 46. This dataset not only reflects the diverse keyboard typing patterns of the participants themselves, but also includes demographic information such as age, gender, hand type, and nationality, providing a rich source for analysis.
- The GREYC-NISLAB dataset [46] includes five different passwords entered by 110 users. The passwords are as follows: "leonardo dicaprio", "the rolling stones", "michael schumacher", "red hot chili peppers", and "united states of america", with the spellings preserved as in the original dataset. In the following text, we will refer to subsets of data using the notations LDC, TRS, MS, RHCP, and USA to denote the data corresponding to each specific password. The number of timestamps (features) ranges from 64 to 92. Each participant entered these passwords ten times with both hands and ten times with the dominant hand, yielding 20 samples per user for each password. Thus, the dataset for each password included 2200 instances, totaling 11,000 data samples for the entire study.

The National Institute of Standards and Technology (NIST) guidelines [47], the leading standard for password security, emphasize that password length contributes more to security than does complexity. The NIST recommends a minimum of eight characters but emphasizes the benefits of longer passwords for increased security. According to the National Cyber Security Centre (NCSC) cyber essentials requirements for IT infrastructure [48], effective password management should include implementing multi-factor authentication or setting a minimum password length. In particular, a minimum password length of at least 12 characters is recommended without a maximum limit.

The keystroke dynamics analyses not only the characters themselves but also detailed temporal data to provide multiple features for each character. In our study, we analyzed seven different passwords with an initial number of features ranging from 31 to 92. Consistent with cybersecurity recommendations for a minimum password length of 12 characters, we standardized the number of features to 37 for all the data by interpolation. The number of features consists of the hold time, release-press time, press-press time, and release-release time of the keystrokes (for more details, see [9,41]). This standardization involved adjusting the number of features of each password — increasing for some and decreasing for others — to a consistent value of 37, thereby organizing the data for processing by the same neural network model.

4. Experiments and results

In this section, we describe a comprehensive evaluation of our proposed methodology for data fusion-based authentication using interpolation methods for keystroke dynamics. We use an integrated dataset approach that combines data from three publicly available sources: the CMU, KeyRecs, and GREYC-NISLAB datasets. These datasets were selected to cover a wide range of typing patterns and behaviors. At the core of our experimental setup is the use of data fusion techniques to standardize keystroke dynamics data across different datasets to a uniform length. Data fusion techniques are represented in particular by interpolation methods, which play a key role in standardizing keystroke dynamics data from different datasets to a uniform length. This standardization is key to creating a deep learning model capable of taking passwords of any length into account, thus enabling a more accurate and universal user authentication system. Our methodology emphasizes the potential of deep learning to capture specific patterns of keystroke dynamics. Through a series of experiments, we aim to demonstrate the effectiveness of our data fusion approach when dealing with passwords of different lengths and thus with different numbers of timestamps. These experiments were designed not only to validate the feasibility of our proposed method but also to identify the most effective strategies for fusing keystroke dynamics datasets. In this way, we aim to demonstrate the enhanced capabilities of our proposed approach to distinguish genuine user actions from potential security threats or insiders, thereby improving the quality of user authentication processes.

As previously mentioned, in our study of user authentication using keystroke dynamics, we unified the data across three keystroke dynamics datasets by using interpolation techniques to standardize the feature set to a fixed length of 37 timestamps for each password. This unification ensures that our models can handle passwords of different lengths while maintaining consistency in the authentication process.

4.1. Experimental design

An Apple MacBook Pro with an M1 Pro chip with a 10-core CPU and a 16-core GPU was used for the computational experiments in our study. This powerful setup, with 32 GB of unified RAM, ensured efficient execution of the complex computational processes required for deep learning-based networks. To implement our models, we used TensorFlow [49], a public library known for its extensive capabilities in large-scale machine learning and analytics. The use of TensorFlow significantly improves the efficiency of our study by allowing us to utilize multiple CPUs and GPUs. This capability has considerably accelerated the data analysis and model training processes, ensuring the timely execution of large-scale computational tasks.

The CNN architecture we used for the SNN, which was designed to process 37×37 input images, is presented in Table 1. Each convolutional layer is followed by batch normalization, dropout, and max pooling operations. This sequence is followed by a flattened layer, whose output is used as input to the dense layer. The final dense layer has 256 outputs. The network output can be considered as an embedding of the original input. Each layer utilizes the Rectified Linear Unit (ReLU) activation function to introduce non-linearity, which enhances the learning ability of the network. The network, which has a depth of 13 layers, covers a total of 2,806,496 parameters.

When separating the dataset for training, validation, and testing a neural network, the goal is to estimate the model's performance as realistically as possible. However, separating data from the same user into training and validation sets can introduce bias and result in overly optimistic performance measures. This is because the model may simply recognize a particular user rather than generalizing the obtained performance. To ensure that the generalization capabilities of the model are accurately estimated, it is important to have different users in the training, validation, and test sets. This separation ensures that the evaluation is based on the model's ability to learn and apply patterns to new, unseen data, which is a more accurate measure of its effectiveness in the real world. Our experiments adopted a 70/15/15 split for training, validation, and testing, respectively. We allocated 15% of the users from each dataset for validation to observe the performance of the network during training. The other 15% of users were allocated for testing and stored separately to evaluate the network's performance on previously unseen data. This separation of data allows us to better simulate real-world conditions when the model encounters data that it did not encounter during training.

When training SNNs with a triplet loss function, an important step is to carefully prepare the triplets of data needed to train the network. Each triplet consists of an anchor sample, which is the reference input sample, a positive sample that is similar to the anchor sample, and a negative sample that is noticeably different from it. Both the anchor and positive samples are taken from the same user, such as images reflecting his/her typing style. The negative sample, however, is taken from another user. This structured learning process instructs the neural network to measure the degree of similarity between the input data and effectively differentiates between the typing patterns of different users. As a result of training, the SNN minimizes the distance between the anchor and positive samples and maximizes the distance between

Table 1
Summary of the convolutional neural network architecture used in the Siamese neural network for keystroke dynamics authentication.

Layers	Output shape	Number of parameters
InputLayer	(None, 37, 37, 3)	0
Conv2D	(None, 37, 37, 32)	1,568
BatchNormalization	(None, 37, 37, 32)	128
Dropout	(None, 37, 37, 32)	0
MaxPooling2D	(None, 18, 18, 32)	0
Conv2D	(None, 18, 18, 64)	18,496
BatchNormalization	(None, 18, 18, 64)	256
Dropout	(None, 18, 18, 64)	0
MaxPooling2D	(None, 9, 9, 64)	0
Flatten	(None, 5,184)	0
Dense	(None, 512)	2,654,720
Dropout	(None, 512)	0
Dense	(None, 256)	131,328

Table 2
Statistical significance (p-values) of different interpolation methods and post-resizing image interpolation methods applied to the CMU and KeyRecs datasets. A p-value derived from ANOVA tests greater than 0.05 indicates no statistically significant difference between the methods.

Dataset	Interpolation	p-value
CMU	Time series interpolation (Linear, Cubic, Nearest)	0.2916
	Post-resizing with interpolation (Bicubic, Bilinear, Nearest)	0.1472
KeyRecs	Time series interpolation (Linear, Cubic, Nearest)	0.1016
	Post-resizing with interpolation (Bicubic, Bilinear, Nearest)	0.0889

the anchor and negative samples. By forcing the network to learn to distinguish between similar (anchor and positive) and dissimilar (anchor and negative) pairs, the model can understand the underlying structure of the data. As a result, similar samples (typing patterns) are closer together, and dissimilar samples are further apart in the feature space (or latent space), making classification or matching easier and more accurate. Triplet learning facilitates the generalization of the model from the concrete examples provided during training to new unseen data. This is because the model learns the concepts of similarities and differences.

4.2. Result validation

In our study, we used ANOVA [50] to assess the statistical significance of the differences in the means of the EERs obtained by the different interpolation methods. This method allows us to determine whether the differences in the EER means obtained are statistically significant or whether they can be explained by chance. The analysis of the differences between the mean EERs obtained by the cubic, linear, and nearest neighbor interpolation methods for the CMU dataset showed a p-value of 0.2916 for the time series data interpolation. Additionally, we assessed the EERs for image post-resizing using interpolation techniques such as bicubic, bilinear, and nearest neighbor methods, which yielded a p-value of 0.1472. Examining the variance of the mean EERs of the KeyRecs dataset obtained by the cubic, linear, and nearest neighbor interpolation methods, the p-value was found to be 0.1016 for the time series interpolation, and 0.0889 for the post-resizing with interpolation. These results (see Table 2), above the usual alpha level of 0.05, indicate that there is no statistically significant difference in the efficiency of the interpolation methods at the 5% significance level. This finding suggests that the observed differences in EER between the interpolation methods can be explained by random variation rather than a definite difference in performance.

Boxplots are commonly used to visually compare the performance of interpolation methods for different data sets. This section presents the performance measures for the test datasets. Figs. 5 and 6 show the EERs for different interpolation methods (linear, cubic, nearest neighbor) applied to time series and for image post-resizing with interpolation (bilinear, bicubic, nearest neighbor) using the CMU and KeyRecs datasets, respectively. Boxplots are often used because they summarize the distribution of data points in a concise way, showing the median, mean, quartiles, and outliers, thus illustrating the tendency and variability of the data. In the boxplot, the center line indicates the median EER for each method. The boundary extends from the first to the third quartile, showing the middle 50% of the data range. The whiskers extend to the farthest points that are not considered outliers. The outliers are shown as individual points outside the whiskers. The triangle indicates the mean EER for each method.

Fig. 5 shows a comparison of how different interpolation methods affect the EER for both time series and post-resizing images. Boxplots for the CMU dataset show the distributions of the EERs for all interpolation methods. Linear interpolation proves to be the most effective method on the CMU dataset, with a mean EER of 0.16462, indicating the high accuracy of this method. On the other hand, the nearest neighbor interpolation method on the post-resizing shows a slightly higher mean EER of 0.17187, and the corresponding cubic interpolation is approximately 0.17634. Notably, linear interpolation maintains not only the lowest mean EER but also the most concentrated interquartile range, indicating a high level of stability of the dataset. In contrast, the cubic and nearest neighbor methods show a larger variation in EER values, indicating less stability in performance. The implications of these findings are also emphasized by the bilinear post-resizing method, which demonstrates a mean EER comparable to that of linear interpolation but with marginally more variation. The smaller range of resizing with the bilinear interpolation method compared to resizing with the bicubic interpolation method suggests that it may be an intermediate option in terms of stability and accuracy. Table 3 supplements these observations by detailing the best, mean, and standard deviation (std) values for accuracy and EER, which provides a more complete understanding of the performance of each method. Lower EER values correlate with higher performance; thus, the low variability of the linear interpolation method confirms its suitability for the task at hand. Although nearest neighbor interpolation competes with resizing using nearest neighbor interpolation, its wider variability indicates its dependence on data specificity. In conclusion, linear interpolation is the most stable and potentially accurate method for the fusion of datasets, while bilinear interpolation can serve as an alternative if moderate variability is acceptable. The key point of the experiments was the best EER value achieved using linear interpolation, which was 0.13672. This value emphasizes the potential of linear interpolation not only for obtaining the most stable mean EER but also for achieving the lowest number of errors in specific cases.

After thoroughly evaluating the interpolation methods applied to the CMU dataset using both the original time series and post-resizing with interpolation, it was concluded that linear interpolation is the most appropriate approach for the efficient fusion of datasets. To further validate this conclusion, we extended our study to the KeyRecs dataset. This set was chosen because of the similarity in password length to the CMU dataset but with a larger feature size. The results of applying different interpolation strategies to the KeyRecs dataset are shown in Fig. 6, and the detailed results are presented in Table 4. Analyzing the post-resizing interpolation results using the KeyRecs dataset shows that the mean EER tends to be slightly lower than that in the case of time series interpolation analysis. Notably, the mean EER for the bilinear and bicubic interpolation methods was slightly greater than 0.19, whereas the nearest neighbor interpolation method showed a mean EER value of 0.18424, although with significant variability. In time series interpolation, the linear interpolation method has consistently lower mean EER values, which is consistent with its performance on the CMU dataset. The bilinear image resizing method shows commendable results. Although its mean EER is slightly greater than that of its nearest neighbor method, its more compact interquartile range indicates reduced variability, which provides more stable performance across different samples. The prominent result of this evaluation was the achievement of the best EER value of 0.15625 using linear

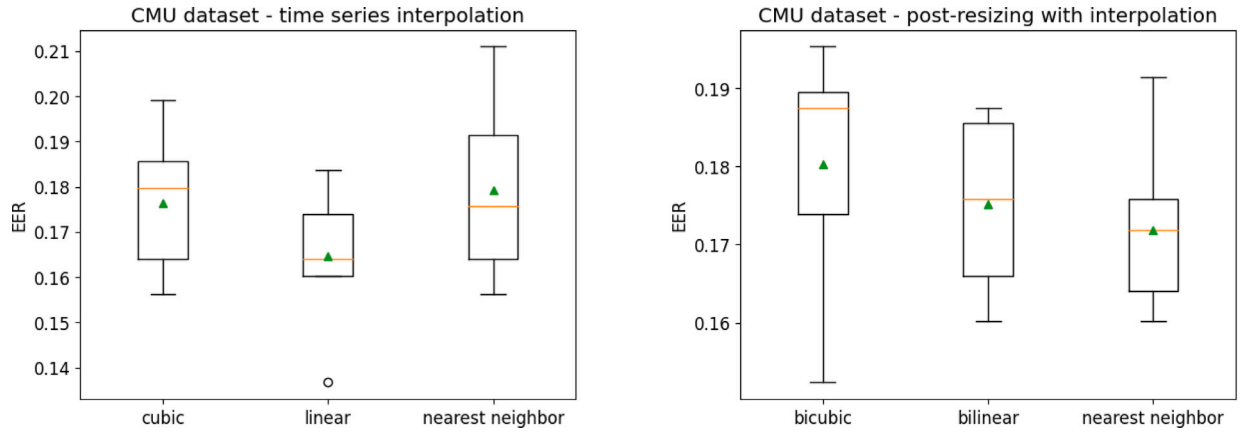

Fig. 5. Comparison of equal error rates for the CMU dataset using different interpolation methods for data standardization: interpolation methods are used to standardize the length of the time series (left); images are post-resized using different interpolation methods (right).

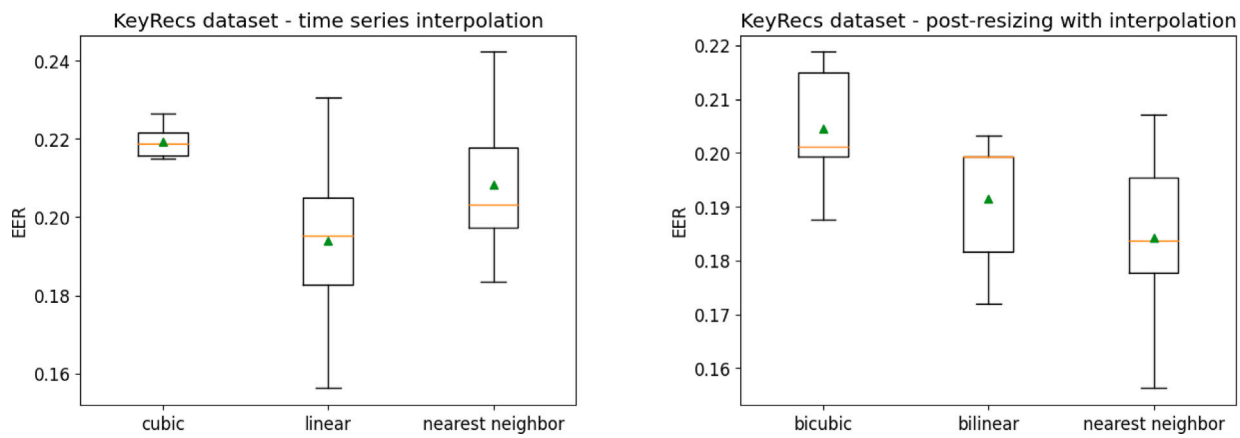

Fig. 6. Comparison of equal error rates for the KeyRecs dataset using different interpolation methods for data standardization: interpolation methods are used to standardize the length of the time series (left); images are postresized using different interpolation methods (right).

Table 3
Performance metrics for the CMU dataset using different interpolation methods for time series standardization and image post-resizing.

Metrics	Time series interpolation			Post-resizing with interpolation		
	Cubic	Nearest neighbor	Linear	Bicubic	Nearest neighbor	Bilinear
Accuracy (best)	0.94141	0.92969	0.91797	0.91406	0.94141	0.92578
Accuracy (mean)	0.91685	0.91574	0.90737	0.90234	0.91016	0.91016
Accuracy (std)	0.01282	0.01062	0.00828	0.00835	0.01235	0.01772
EER (best)	0.15625	0.15625	0.13672	0.15234	0.16016	0.16016
EER (mean)	0.17634	0.17913	0.16462	0.18025	0.17187	0.17188
EER (std)	0.01423	0.02007	0.01423	0.01376	0.01023	0.02738

Table 4
Performance metrics for the KeyRecs dataset using different interpolation techniques for data fusion.

Metrics	Time series interpolation			Post-resizing with interpolation		
	Cubic	Nearest neighbor	Linear	Bicubic	Nearest neighbor	Bilinear
Accuracy (best)	0.91406	0.90625	0.91406	0.89844	0.91016	0.90625
Accuracy (mean)	0.87956	0.89388	0.89779	0.88346	0.89388	0.89388
Accuracy (std)	0.01648	0.01264	0.01569	0.01159	0.01137	0.00941
EER (best)	0.21484	0.18359	0.15625	0.18750	0.15625	0.17188
EER (mean)	0.21940	0.20833	0.19401	0.20443	0.18424	0.19141
EER (std)	0.00417	0.01909	0.02296	0.01120	0.01633	0.01256

interpolation on the KeyRecs dataset, which is an indication of the method's potential to provide strong user authentication on different datasets.

A comparative evaluation of the CMU and KeyRecs datasets revealed a definite relationship between the different interpolation strategies and the characteristics of each dataset, particularly in terms of accuracy and EER. This analysis shows that although the performance of the interpolation methods varies depending on the specifics of the dataset, linear interpolation consistently maintains high stability in all cases. This stable EER across different datasets emphasizes the effectiveness of linear interpolation and makes it the preferred data fusion method for training neural network models. This is particularly

Table 5
Performance metrics using the GREYC-NISLAB datasets using linear interpolation. The best, mean and standard deviations of equal error rates and accuracy values by specific passwords ("leonardo dicaprio", "michael schumacher", "red hot chili peppers", "the rolling stones" and "united states of america") are presented, demonstrating the effectiveness of linear interpolation when dealing with passwords of different lengths for user authentication.

Metrics	Datasets				
	LDC	MS	RHCP	TRS	USA
Accuracy (best)	0.93750	0.96875	1.00000	0.96875	0.90625
Accuracy (mean)	0.87500	0.93229	0.87500	0.89583	0.83854
Accuracy (std)	0.03608	0.04199	0.06250	0.03898	0.04199
EER (best)	0.15625	0.15625	0.15625	0.1250	0.15625
EER (mean)	0.19792	0.18750	0.21354	0.17708	0.22396
EER (std)	0.02329	0.02552	0.04199	0.02946	0.03335

relevant in scenarios requiring improved accuracy and reduced EER in different data contexts.

The analysis of the GREYC-NISLAB dataset presented in Fig. 7 and Table 5 shows that the linear interpolation method applied to keystroke dynamics data with different initial feature lengths is quite promising. The GREYC-NISLAB dataset consisted of five different passwords, each corresponding to a significant person or phrase: "leonardo di caprio" (LDC), "michael shumacher" (MS), "red hot chili pepper" (RHCP), "the rolling stones" (TRS) and "united states of america" (USA). This uniform feature transformation to 37 for a wide range of passwords demonstrates the potential of a generalized authentication solution capable of handling passwords of different lengths with significant efficiency. Notably, the obtained EERs for different passphrases indicate that the linear interpolation method performs well even after reducing the feature sizes, providing stable authentication performance. For example, the passwords associated with LDC and MS yielded mean EER values of 0.19792 and 0.18750, respectively, with relatively small standard deviations. This consistency indicates the ability of the method to retain the distinctive features of the set after feature compression, which is important for accurate user authentication. The highest variability of the password corresponding to the USA, which also had the longest initial feature length, raises the question of the scalability of the method when dealing with significantly long passwords. However, even in this case, the mean EER did not exceed 0.22396, which is relatively moderate and indicates some robustness of the method. Moreover, the better accuracy rates, especially for LDC and MS, confirm the potential of this approach. These performances indicate that even when the feature size is compressed, the important dynamics of individual keystrokes are largely preserved, which is promising for real-world applications where the password length can vary significantly. The most compelling aspect of this analysis is the best EER values achieved for the different passwords. Notably, the passwords associated with LDC, MS, RHCP, and USA achieved the best EER value of 0.15625, which is exceptionally low. This low EER is indicative of the high accuracy of the system, where false accepts and false rejects are minimized, thus providing a high degree of confidence in the authentication process. For TRS, the best EER is even lower at 0.1250, further emphasizing the potential of this approach. These best EERs are particularly important because they demonstrate the capabilities of the model. When considering such a system for critical infrastructure security, where accuracy is extremely important, these best EER values provide strong evidence that the method is capable of meeting stringent security requirements.

The effectiveness of this approach in handling passwords of different lengths and maintaining a relatively low and stable EER for different passwords is significant. This suggests that such a methodology can be applied to develop a versatile authentication system that is not only secure but also adaptable to natural changes in password lengths that occur in real-world cybersecurity scenarios.

For further experiments aimed at improving the performance of data fusion for training a single neural network for passwords of any length, three different experiments were designed, taking into account previous insights into the variability of performance with passphrase length:

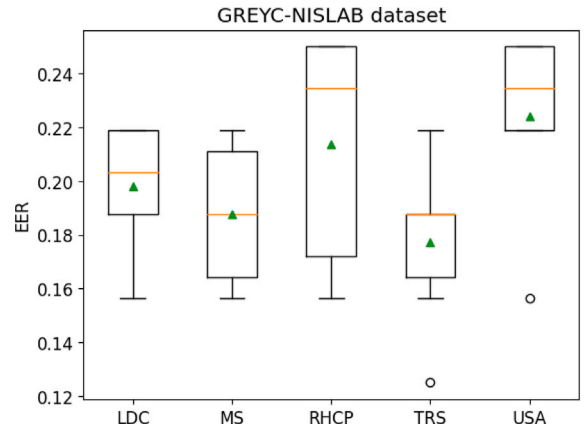

Fig. 7. Comparison of equal error rates for all passwords in the GREYC-NISLAB dataset using linear interpolation, demonstrating the effectiveness of the proposed methodology when dealing with passwords of different lengths for user authentication.

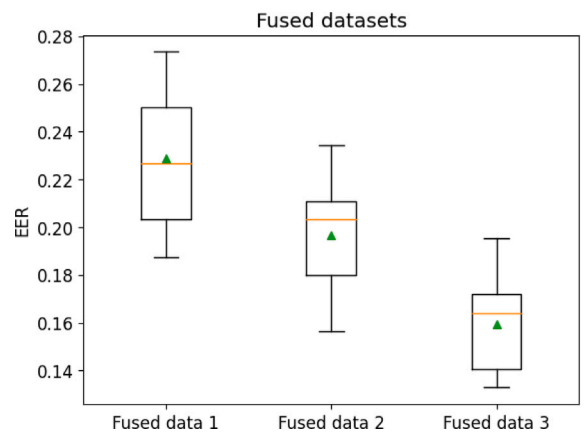

Fig. 8. Comparison of equal error rates for fused keystroke dynamics datasets using different data fusion strategies, showing the impact on model accuracy and generalization to unseen data.

- In the first experiment, the passphrases from the KeyRecs, CMU, and GREYC-NISLAB datasets were fused into a single dataset to train the SNN model. The trained model was then evaluated on an unseen subset of samples consisting of 15% of the users from each of these datasets. This combined dataset is referred to as "Fused data 1".
- The second experiment was designed to address the observed performance problems associated with longer passphrases. Here, the KeyRecs and GREYC-NISLAB datasets were fused, explicitly excluding the longer passphrases of RHCP and USA, as it was acknowledged that they had underperformed in previous analyses. The network was subsequently tested on the full CMU dataset. This combined dataset is referred to as "Fused data 2".
- In the third experiment, the fusion of the KeyRecs, CMU, and GREYC-NISLAB datasets also excluded the RHCP and USA passphrases, on the assumption that removing the longer passphrases might improve overall performance. This newly trained network was then tested with an unseen subset of 15% of the users from the CMU dataset, providing a reliable assessment of the model's generalizability. This combined dataset is referred to as "Fused data 3".

When analyzing "Fused data 1", which consists of passwords of various lengths, the best EER is 0.1875 (see Fig. 8). Using "Fused data 2", there is a noticeable improvement in the EER, suggesting that the elimination of long passwords may have had a positive impact

on the model's performance. The use of "Fused data 3" achieves a lower mean EER and a narrower interquartile range. In particular, "Fused data 3" has the best EER value of 0.13281, which demonstrates a remarkable level of accuracy in authenticating users. This shows robust and stable performance under different user data and indicates successful improvement in the neural network's ability to generalize on unseen data.

5. Discussion

This study presents a keystroke dynamics-based authentication methodology focused on data fusion and deep learning techniques. To the best of our knowledge, this study is the first to use a standardized password length across multiple datasets in combination with a data fusion approach for keystroke dynamics-based authentication. Due to the specific goals and experimental conditions of our study, direct comparison with previous works is challenging. Our results differ from those of other authors (see [4,10,22,51–53]) because our aim was to standardize the number of password features to a commonly used length and to train a single network suitable for all passwords. While our approach is promising, especially in improving the generalizability of authentication systems for different user groups and is not dependent on password length, it is essential to consider how our results relate to existing research in this area.

5.1. Comparison with previous studies

The performance of keystroke biometrics is affected by variables such as emotional state, body posture, keyboard type, and other situational factors [7]. For some physiological biometrics, such as fingerprints or iris recognition, the EER can range from 0.001 to 0.0077 [54]. While for behavioral biometrics, the EER can range from 0.1 to 0.2, which is often considered appropriate to ensure a balance between user experience and security measures [10,52,53]. In this study and experiments, we presented a methodology for user authentication based on keystroke dynamics, which differs in that the testing phase uses only data from new, unseen users. Unlike previous studies [4,22,51] in which the separation of training and test samples based on the same users could introduce bias and overestimate the results of performance evaluation, in our model, performance was evaluated only on new, unseen users. By choosing different sets of users for training and testing, we ensure that our results reflect the ability of the model to generalize across different behaviors and typing patterns, which is important for real-world applications. This achievement is noteworthy given the inherent variability and complexity of biometric datasets, which often pose significant challenges for pattern recognition and anomaly detection systems. The methodology developed in this research provides a solid basis for improving neural network training to enhance the security and accuracy of biometric authentication systems.

A number of studies have reported lower EERs than those observed in our work, which requires further discussion of methodological differences and their implications for real-world applications. For example, in the study [51], using a CNN-based approach on the GREYC-NISLAB dataset, the best obtained EER was approximately 0.05. However, it is important to note the methodological differences on which the obtained result depends. In this study, combining multiple passwords into a single long password consisting of 99 characters and 376 attributes makes it possible to obtain an extended set of features that affects the performance of the model. However, this approach cannot always be directly applied to real-world scenarios, where users typically enter shorter and fixed passwords. In contrast, in our research, the password length is standardized to 37 features, which, although limiting the feature set, provides a more realistic and practical approach to user authentication.

Similar studies such as [55,56] have demonstrated the effectiveness of random forest and support vector machines algorithms in keystroke dynamics, especially in scenarios with free-text input or long password phrases. These cases utilize a broader set of features available in free-text scenarios, which improves the performance of the models. In contrast, our work focuses on fixed-text authentication, where the feature set is restricted to a standard password length. This limitation, while challenging, is crucial for developing a generalized model that can handle a variety of user inputs without requiring significant customization for each use case.

5.2. Justification for data fusion

The use of data fusion in our study is motivated by the effort to create a single unified model capable of handling different types of passwords across a wide user database. Training separate models for each user or dataset is not only resource intensive, but also impractical for large-scale applications, especially in critical infrastructure where efficiency and scalability are crucial.

Our data fusion approach, which combines multiple datasets into a standardized format, addresses these challenges by creating a more flexible model. This model can generalize to different user profiles, making it applicable in real-world applications where users may have different typing behaviors. The slightly higher EER observed in our study reflects the rigorous testing conditions in which the model was evaluated on unseen data from new users, in contrast to some previous studies in which testing was performed on validation data from known users, potentially leading to lower EER due to over-fitting.

6. Conclusions and future work

At a time when cybersecurity threats are becoming increasingly complex and widespread, protecting critical infrastructure has never been more important. The resilience of critical systems, including financial services, military installations, healthcare, and utilities, to cyberthreats is critical to national security, economic stability, and public safety. With this background, our research takes on special importance by proposing an advanced method that combines deep learning and data fusion to improve user authentication by analyzing keystroke dynamics. Our methodology consists of a data fusion process that transforms keystroke data from disparate datasets into a uniform format, enabling the development of a deep learning model that recognizes users by analyzing the unique rhythm and timing of keystrokes, regardless of password length. An important aspect of our study is the application of the GAFMAT (GAbor Filter MAtrix Transformation) application, which enables the transformation of keystroke timing data into a form suitable for a Siamese neural network with convolutional neural network branches, which allows the model to be trained efficiently based on the complex patterns found in keystroke dynamics.

Our work utilized data integration from three key datasets: the CMU dataset, the KeyRecs dataset, and the GREYC-NISLAB dataset. This integration resulted in an extensive fused dataset that significantly improved the generalizability of the model and its adaptability to a wide range of user behaviors. Two types of data fusion strategies were used in the study: time series interpolation and image post-resizing with interpolation. The application of ANOVA provided a statistical basis for evaluating the effectiveness of different interpolation methods. Our analysis yielded p-values exceeding the 0.05 threshold, indicating that there is no statistically significant difference in the effectiveness of the interpolation methods at the usual significance level. The results suggest that linear interpolation offers a balanced approach, yielding the lowest mean EER and showing stable performance across different datasets.

By utilizing specific interpolation-based data fusion strategies, as well as a Siamese neural network with a triplet loss function, the best equal error rate of 0.13281 was achieved for the unseen fused data. This indicates that our method can extend the capabilities of

user authentication systems, thereby providing more robust security measures for critical infrastructures and insider detection.

Despite the encouraging results, the study has certain limitations. In the future, we intend to examine additional datasets and refine our data fusion strategies to increase the precision of the authentication process. A potential direction for future research is to implement the proposed methodology in free-text keystroke dynamics, which could lead to more scalable and user-friendly authentication methods. In addition, it may be beneficial to consider integrating other interpolation techniques, such as the dynamic time warping approach and Fourier transform. Future work will also aim to incorporate quantile transformation as a preprocessing method. This could potentially improve the performance of authentication models.

CRediT authorship contribution statement

Arnoldas Budžys: Writing – review & editing, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Olga Kurasova:** Writing – review & editing, Writing – original draft, Validation, Data curation, Conceptualization. **Viktor Medvedev:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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